Engineering study notes

Aircraft Performance

Theory, Equations and Worked Calculations

A study guide for the V7 project

> Learn the physical ideas first, derive the main equations, then apply them to representative calculations and explain their limitations. This guide covers the engineering and numerical concepts, not Python syntax.

The calculation chain

Units and conditions → density and wind → forces and speeds

Ground roll and climb → takeoff or landing distance → runway comparison.

Based on V7 equations and constants. Values are generated by `make_values.py`. Hand calculations may differ slightly due to rounding.

November 2026

Educational model; not approved flight-planning data.

Contents

1 How to study this project 2
2 Symbols, units and signs 3
3 Atmosphere: ISA, temperature and density 4
4 TAS, GS and wind components 6
5 Lift, drag and the drag polar 8
6 Stall speed and the V-speed schedule 10
7 Propeller power, thrust and turbocharging 12
8 Jet thrust, Mach number and flat rating 13
9 Ground roll: Newton's second law 15
10 Numerical integration: why distance is not one division 17
11 Rotation, climb and transition to screen height 19
12 Balanced field length and the AEO limit 21
13 Reading the BFL chart 23
14 Landing distance and landing-distance factors 24
15 Runway checks, margins and sensitivity 26
16 A complete C172 calculation to explain aloud 27
17 What changes what, and why 28
18 Practice questions: try these without the answers 30
19 Worked answers 31
20 Questions you should be able to answer aloud 32
21 Compact equation checklist 33
22 Textbook reading map 34
References 35

How to read the citations. The calculations and numerical choices in this guide follow V7. Textbook citations identify the underlying theory and suggested reading; they do not establish that each V7 approximation appears verbatim in a book. Chapter and section pointers for Mair & Birdsall and Filippone were checked against publisher contents lists. No unverified page or equation numbers are given.

1 How to study this project

For each equation, practise five things: identify the output, define every symbol and unit, explain the physical reason for each term, substitute numbers, and predict how the answer changes when an input changes.

Study block	What you should be able to do without notes
1. Foundations	Convert units; distinguish mass, weight, TAS and GS.
2. Atmosphere	Calculate ISA deviation, pressure, density and speed of sound.
3. Aerodynamics	Calculate lift and drag; derive stall speed from $L = W$.
4. Propulsion	Explain the propeller thrust cap, turbocharging, flat rating and OEI.
5. Flight segments	Write the accelerating and braking force balances; assemble takeoff and landing distances.
6. Field length	Explain the BFL crossing, AEO factor and runway margins.
7. Defence	Explain sensitivity, calibration, numerical integration and limitations.

Three kinds of equation appear here.

- **Physical relations:** Newton's second law, the ideal-gas relation, and lift/drag expressed using aerodynamic coefficients.
- **Model approximations:** constant coefficients, fixed V-speed ratios, constant propeller efficiency and a circular transition arc.
- **Project factors:** wind credits, the AEO distance multiplier and landing-distance factors. Learn how this project applies them without claiming that it implements every certification requirement.

> Being able to repeat a formula is the starting point. Being able to explain why the project uses it, what is held constant, and where it can fail is the goal.

A useful routine: read one section, reproduce its equations on blank paper, do its worked calculation, then answer the check question aloud. Complete the mixed practice set only after working through the full calculation chain. Each added theory page belongs to the preceding topic: use it to explain why the equations work before repeating the numerical example.

2 Symbols, units and signs

Symbol	Meaning	SI unit
m, W	Aircraft mass; aircraft weight	kg; N
T, D, L	Thrust, drag, lift	N
Θ, p, ρ	Absolute air temperature, pressure, density	K; Pa; kg/m^3
V, V_{GS}	TAS; ground speed along the runway	m/s
V_w, V_x	Headwind; crosswind component	m/s or kt
S, q	Reference wing area; dynamic pressure	m^2; Pa
C_L, C_D, k	Lift, drag and induced-drag coefficients	dimensionless
μ, θ	Friction coefficient; runway slope angle	dimensionless; rad
γ, n	Climb angle; transition load factor	rad; dimensionless
P, η_p	Shaft power; propeller efficiency	W; dimensionless
a, a_{GS}	Speed of sound; ground acceleration	m/s; m/s^2

We use Θ for temperature to avoid confusing it with thrust T. Pressure altitude h is measured above the atmospheric reference; screen height is measured above the runway.

$$W = mg_0, \qquad g_0 = 9.80665\,\mathrm{m/s^2},$$
$$1\,\mathrm{ft} = 0.3048\,\mathrm{m}, \qquad 1\,\mathrm{kt} = 1852/3600\,\mathrm{m/s},$$
$$1\,\mathrm{lb} = 0.45359237\,\mathrm{kg}, \qquad 1\,\mathrm{lbf} = 4.4482216152605\,\mathrm{N},$$
$$1\,\mathrm{hp} = 745.69987158227\,\mathrm{W}, \qquad \Theta[\mathrm{K}] = \Theta[^\circ\mathrm{C}] + 273.15.$$

Area conversions require the *square* of the length factor: $1\,\mathrm{ft}^2 = 0.3048^2\,\mathrm{m}^2$. A temperature difference of 20°C equals 20 K; an absolute temperature of 20°C equals 293.15 K.

Signs: positive headwind opposes aircraft motion; tailwind is negative. Positive slope is uphill in the direction of operation. Positive crosswind is from the right. Convert degree inputs to radians before trigonometric evaluation.

Example: $2550\,\mathrm{lb} \times 0.45359237 = 1,156.66\,\mathrm{kg}$, then $W = 1,156.66 \times 9.80665 = 11,342.97\,\mathrm{N}$. The number in kg is not a force.

Check yourself. Why is multiplying pounds by g_0 directly wrong? Because lb must first be converted to kg if g_0 is in m/s^2 and the required force is N.

3 Atmosphere: ISA, temperature and density

Theory and further reading. Anderson, *Introduction to Flight* [2]; Filippone, Sec. 8.1 [4].

The project uses a constant-lapse-rate troposphere, valid for $0 \le h \le 11000\,\mathrm{m}$.

$$\begin{aligned}
\Theta_{\mathrm{std}} &= 288.15 - 0.0065h, \\
p &= 101325\left(\frac{\Theta_{\mathrm{std}}}{288.15}\right)^{g_0/(0.0065R)}, \\
\Theta &= \Theta_{\mathrm{std}} + \Delta\Theta_{\mathrm{ISA}}, \\
\rho &= \frac{p}{R\Theta}, \qquad R = 287.05\,\mathrm{J/(kg\,K)}, \\
\sigma &= \frac{\rho}{1.225}, \qquad a = \sqrt{1.4R\Theta}.
\end{aligned}$$

The pressure expression follows from hydrostatic balance $dp/dh = -\rho g_0$, the ideal-gas relation and the standard temperature lapse. An ISA deviation changes the actual temperature and density, but the pressure at a specified *pressure altitude* still follows the standard profile.

Given OAT in °C:

$$\Delta\Theta_{\mathrm{ISA}} = \mathrm{OAT} - (\Theta_{\mathrm{std}} - 273.15).$$

Worked example: 5000 ft, ISA+20.

$$\begin{aligned}
h &= 5000(0.3048) = 1524\,\mathrm{m}, \\
\Theta_{\mathrm{std}} &= 288.15 - 0.0065(1524) = 278.244\,\mathrm{K}, \\
\Theta &= 278.244 + 20 = 298.24\,\mathrm{K}, \\
p &= 84,307.11\,\mathrm{Pa}, \quad \rho = 0.9848\,\mathrm{kg/m^3}, \quad \sigma = 0.8039.
\end{aligned}$$

> At fixed pressure altitude, hotter air has lower density because $\rho = p/(R\Theta)$. Lower density raises the TAS needed for a given lift. It can also reduce available power or thrust, depending on the engine model.

Pressure altitude versus density altitude: pressure altitude represents the pressure condition. Density altitude is the ISA altitude corresponding to the actual density. This project calculates density directly; it does not need a separate density-altitude conversion.

Check yourself. Does ISA+20 mean the OAT is 20°C? No. It means 20 degrees above standard temperature at that altitude.

3 Theory: why pressure and density fall with height

Theory and further reading. Atmosphere background: Anderson [2]; Filippone, Sec. 8.1 [4]. The derivation below uses the V7 constants.

Start with a stationary layer of air. The pressure force from below must balance the pressure force from above and the layer's weight. For a thin layer of thickness dh, the balance gives

$$\frac{dp}{dh} = -\rho g_0.$$

The negative sign means pressure falls as height increases. Pressure is force per area; density is mass per volume. They are related, but they are not the same property. Combining hydrostatic balance with the ideal-gas relation $p = \rho R\Theta$ gives $dp/p = -g_0\, dh/(R\Theta)$.

Where the ISA pressure exponent comes from. In the troposphere, assume a constant temperature lapse rate λ: $\Theta_{\text{std}} = \Theta_0 - \lambda h$. Integrating from sea level to h gives

$$\ln \frac{p}{p_0} = \frac{g_0}{R\lambda} \ln \frac{\Theta_0 - \lambda h}{\Theta_0}, \qquad p = p_0 \left(\frac{\Theta_{\text{std}}}{\Theta_0} \right)^{g_0/(R\lambda)}.$$

The power law is therefore the result of hydrostatic balance, an ideal gas, a linear temperature profile and constant gravity. It is not a universal pressure law for every height or weather condition.

Why a hot day changes density without changing pressure here. V7 takes *pressure altitude* as an input: it identifies a pressure on the standard profile. At that fixed pressure, applying ISA deviation changes Θ, so $\rho = p/(R\Theta)$ decreases as temperature increases. At the same pressure, a change from 280 K to 300 K gives $\rho_{300}/\rho_{280} = 280/300 = 0.9333$: about 6.7% less density. This does not say pressure at a fixed geographical elevation never changes.

Why the aircraft cares. Lower density reduces aerodynamic forces at a given TAS, so more TAS is needed to generate a specified lift. It can also reduce available engine output. The atmosphere therefore changes both the target speed and the acceleration available to reach it. Speed of sound depends on absolute temperature through $a = \sqrt{\gamma_{\text{air}} R\Theta}$; Mach number is V/a.

> **Explain aloud:** pressure altitude identifies pressure; ISA deviation sets the temperature departure; together they determine density. Density altitude is the altitude in ISA that would have this same density, not another name for the input pressure altitude.

4 TAS, GS and wind components

Theory and further reading. Mair & Birdsall, Ch. 2 [3]; Filippone, Sec. 8.2 [4]. The wind-credit factors below describe V7.

Aerodynamic forces depend on motion relative to air. Runway distance depends on motion relative to the ground. For the project's along-runway model:

$$V_{\mathrm{TAS}} = V_{\mathrm{GS}} + V_w, \qquad V_{\mathrm{GS}} = V_{\mathrm{TAS}} - V_w.$$

At a fixed liftoff TAS, a headwind reduces the ground speed that must be reached. For example, 70 KTAS with a 10 kt headwind corresponds to 60 kt GS; with a 10 kt tailwind, it corresponds to 80 kt GS.

For wind *from* heading ψ_w and runway true heading ψ_r:

$$V_w = V_{\mathrm{wind}} \cos(\psi_w - \psi_r),$$
$$V_x = V_{\mathrm{wind}} \sin(\psi_w - \psi_r).$$

Example: wind from 030° at 20 kt, runway heading 000°: $V_w = 20\cos 30^\circ = 17.32$ kt and $V_x = 20 \sin 30^\circ = 10.00$ kt from the right.

The runway-analysis calculation applies:

$$V_{w,\mathrm{perf}} = \begin{cases} 0.5 V_w, & V_w \geq 0, \\ 1.5 V_w, & V_w < 0. \end{cases}$$

Thus a reported 10 kt headwind becomes 5 kt, while a 10 kt tailwind becomes −15 kt. These factors reduce the credited benefit and increase the penalty.

> Use the factored wind once. A direct takeoff, landing or BFL calculation uses the headwind supplied to it; runway analysis factors reported wind before calling those calculations. Factoring it again would change the intended model.

IAS/CAS/TAS: the project reports TAS, labelled KTAS in knots. IAS is an instrument indication; CAS corrects instrument and position error. TAS is actual speed through the air. Do not compare a computed KTAS directly with an AFM KCAS schedule without the appropriate conversion.

Crosswind is reported but does not directly change this one-dimensional force balance, nor does the project enforce an aircraft-specific crosswind limit.

Check yourself. At brake release in a headwind, must aerodynamic lift be zero? No. GS is zero, but TAS equals the headwind in this model.

4 Theory: two reference frames, one aircraft

Theory and further reading. Flight-mechanics background: Mair & Birdsall, Ch. 2 [3].

Velocity is always relative to something. An aerodynamic force depends on the aircraft's motion relative to the air. Runway distance depends on its motion relative to the ground. Along the runway, with positive V_w defined as headwind,

$$V_{\mathrm{TAS}} = V_{\mathrm{GS}} + V_w, \qquad \frac{ds}{dt} = V_{\mathrm{GS}} = V_{\mathrm{TAS}} - V_w.$$

The air does not need the aircraft to move over the ground before it can exert a load. A stationary aircraft in a headwind already has airflow over it.

Separate a force calculation from a distance calculation. Suppose liftoff needs 60 kt TAS and the headwind is 10 kt. The aircraft reaches that TAS at 50 kt GS. The lift requirement is unchanged if density, weight and configuration are unchanged, but the aircraft travels over the ground more slowly. A 10 kt tailwind instead requires 70 kt GS for the same TAS.

For a deliberately simplified comparison, suppose each case accelerated from rest at the same constant ground acceleration. Then $s \propto V_{\mathrm{GS}}^2$:

$$\frac{s_{\mathrm{head}}}{s_{\mathrm{calm}}} = \left(\frac{50}{60}\right)^2 = 0.694, \qquad \frac{s_{\mathrm{tail}}}{s_{\mathrm{calm}}} = \left(\frac{70}{60}\right)^2 = 1.361.$$

These ratios explain the strong sensitivity to wind. They are not V7 predictions: in V7 the force history also changes because TAS differs at each GS.

Why a wind component uses a cosine. Wind is a vector. Projecting it onto the runway gives $V_w = V\cos\Delta$; the perpendicular projection gives crosswind $V_x = V\sin\Delta$. The heading convention must use the direction the wind comes from consistently. A 90-degree crosswind has no along-runway component, even though it can still matter greatly for controllability. V7's one-dimensional force model does not simulate lateral tyre forces or rudder control.

A steady-wind assumption is hidden in the acceleration equation. If V_w is constant, differentiating the velocity relation gives $dV_{\mathrm{TAS}}/dt = dV_{\mathrm{GS}}/dt$. In a changing wind, an additional dV_w/dt term appears. The project is not a wind-shear or gust-response simulation.

> **Explain aloud:** use TAS in lift, drag and propulsion; use GS to accumulate runway distance. A headwind does not reduce the required lift or the aircraft's mass. It changes the relation between airspeed and ground motion.

5 Lift, drag and the drag polar

Theory and further reading. Anderson [1]; Mair & Birdsall, Chs. 1–3 [3].

$$
\begin{aligned}
q &= \tfrac{1}{2}\rho V_{\mathrm{TAS}}^2, \\
L &= qSC_L, \\
C_D &= C_{D_0} + kC_L^2, \\
D &= qSC_D = qS(C_{D_0} + kC_L^2).
\end{aligned}
$$

q is dynamic pressure, not force. Multiplying it by area gives a force scale; the dimensionless coefficient describes how much of that scale becomes lift or drag. C_{D_0} represents zero-lift drag in the chosen configuration. The kC_L^2 term represents induced drag. These coefficients are held fixed or selected by flight segment in this project; they are not a complete aerodynamic database.

Worked calculation. With $\rho = 1.225$, $V = 40\,\mathrm{m/s}$, $S = 16.2\,\mathrm{m}^2$, $C_L = 0.4$, $C_{D_0} = 0.032$ and $k = 0.045$:

$$
\begin{aligned}
q &= \tfrac{1}{2}(1.225)(40)^2 = 980\,\mathrm{Pa}, \\
L &= 980(16.2)(0.4) = 6350.4\,\mathrm{N}, \\
C_D &= 0.032 + 0.045(0.4)^2 = 0.0392, \\
D &= 980(16.2)(0.0392) = 622.34\,\mathrm{N}.
\end{aligned}
$$

Why induced drag is often said to fall with speed. At *fixed* C_L, both lift and drag above scale with V^2. But in approximately level flight, $L = W$ requires $C_L = W/(qS)$. Substituting gives:

$$
D = qSC_{D_0} + \frac{kW^2}{qS}.
$$

The first term grows with speed; the second falls with speed. Always state what is held fixed when discussing a trend.

Useful extension: for a simple finite-wing approximation, $k \approx 1/(\pi e\, AR)$, where $AR = b^2/S$ and e is an efficiency factor. V7 takes k as an input rather than calculating it from span and e.

Check yourself. Does doubling TAS always quadruple drag? Only if density, area and C_D remain fixed. Maintaining the same lift changes C_L and induced drag.

5 Theory: why drag has two competing parts

Theory and further reading. Anderson [1]; Mair & Birdsall, Chs. 1–3 [3]. See also NASA's induced-drag explanation.

What dynamic pressure means. The quantity $q = \rho V^2/2$ has units of pressure and provides a scale for aerodynamic loading. Multiplying by area gives a force scale; C_L and C_D express how the shape, attitude and flow turn that scale into lift and drag. The equations do not determine the coefficients themselves. Those require data or a separate aerodynamic model.

Why producing lift also produces drag. A finite wing deflects the surrounding airflow. Downwash changes the direction of the aerodynamic force, giving a component opposing the flight direction. A common finite-wing model is

$$C_{D_i} = \frac{C_L^2}{\pi e\, AR} = kC_L^2, \qquad AR = \frac{b^2}{S},$$

where b is span and e represents departures from an ideal spanwise loading and other effects in the adopted polar. V7 supplies k directly. Its C_{D_0} groups the modeled zero-lift drag contributions.

The key distinction: fixed lift versus fixed lift coefficient. For steady level flight, set $L = W$, hence $C_L = W/(qS)$. Substituting this into the parabolic polar gives

$$D = qSC_{D_0} + \frac{kW^2}{qS} = \underbrace{\tfrac{1}{2}\rho S C_{D_0} V^2}_{\text{zero-lift term}} + \underbrace{\frac{2kW^2}{\rho S V^2}}_{\text{induced term}} .$$

At fixed weight and density, the first term rises with speed while the second falls. Their sum has a minimum. Differentiating with respect to q shows that minimum drag occurs when the two terms are equal, giving $C_L = \sqrt{C_{D_0}/k}$ within this idealized polar.

Why the ground-roll model behaves differently. V7 holds $C_{L,\text{gnd}}$ fixed during the roll. Therefore both modeled drag terms grow with q, and lift grows with q too. Do not use the level-flight claim "induced drag falls as speed rises" for a calculation holding C_L fixed. Always state what remains constant when describing a trend.

> **Explain aloud:** the familiar U-shaped drag curve assumes lift is held near weight. The aircraft changes C_L as speed changes. A fixed-attitude runway calculation makes a different assumption.

6 Stall speed and the V-speed schedule

Theory and further reading. Anderson [1]; Filippone, Secs. 8.2 and 9.6 [4]. Fixed speed ratios are V7 assumptions.

At the stall limit in the project's one-g approximation:

$$W = L = \tfrac{1}{2}\rho V_S^2 S C_{L,\max} \quad \Longrightarrow \quad \boxed{V_S = \sqrt{\frac{2W}{\rho S C_{L,\max}}}}.$$

Stall is associated with reaching the limiting lift coefficient, normally tied to excessive angle of attack; the quoted stall speed depends on the conditions. V_{S1} uses takeoff flaps; V_{S0} uses landing flaps here.

Speed	Meaning in this model	Schedule
V_R	Rotation speed	$1.1V_{S1}$
V_2	Initial climb speed	$1.2V_{S1}$
V_{TR}	Transition-arc speed	$1.15V_{S1}$
V_{REF}	Landing approach speed	$1.3V_{S0}$
V_{TD}	Touchdown speed	$1.15V_{S0}$
V_1	Engine-failure/decision-speed trial	Solved from go/stop distances

Scaling you should know:

$$V_S \propto \sqrt{W}, \qquad V_S \propto \rho^{-1/2}, \qquad V_S \propto C_{L,\max}^{-1/2}.$$

Increasing weight by 10% raises stall speed by $\sqrt{1.10} - 1 = 4.88\%$. Reducing density to 80% raises stall TAS by $1/\sqrt{0.8} - 1 = 11.80\%$. Neither percentage is the same as the eventual takeoff-distance change.

C172 example: using $W = 11,342.97\,\mathrm{N}$, $S = 16.2\,\mathrm{m}^2$, $C_{L,\max,TO} = 1.5$ and the model's sea-level density:

$$V_{S1} = 27.61\,\mathrm{m/s}, \quad V_R = 30.37\,\mathrm{m/s} = 59.03\,\mathrm{KTAS}, \quad V_2 = 33.13\,\mathrm{m/s}.$$

> The fixed multipliers are project assumptions, not a universal certified V-speed schedule. The model does not independently calculate VMCG, VMU or VLOF.

Check yourself. Does a 20% increase in $C_{L,\max}$ reduce stall speed by 20%? No: the speed factor is $1/\sqrt{1.2} = 0.9129$, a reduction of 8.71%.

6 Theory: stall is a lift limit, not an arbitrary speed

Theory and further reading. Anderson [1, 2]; Filippone, Sec. 8.2 [4].

Begin with angle of attack. Angle of attack is the angle between the wing's reference chord and the relative airflow. Increasing it generally raises C_L over a useful range. Beyond a critical region, substantial flow separation limits the lift coefficient. The peak value is represented by $C_{L,\max}$ for a given configuration. Stall is therefore tied to the aerodynamic state, not one universal indicated or true airspeed.

Derive the speed from the required lift. For unaccelerated level flight, or the project's simplified lift-off threshold, require $L = W$ and set $C_L = C_{L,\max}$:

$$W = \tfrac{1}{2}\rho V_S^2 S C_{L,\max} \quad \Longrightarrow \quad V_S = \sqrt{\frac{2W}{\rho S C_{L,\max}}}.$$

This is TAS when local density is used. It assumes a known maximum lift coefficient and ignores detailed ground effect, transient pitch dynamics and configuration changes during rotation.

Use ratios to understand the square root. With everything else fixed, 10% more weight raises stall speed by $\sqrt{1.10} - 1 = 4.88\%$. A 20% reduction in density raises it by $1/\sqrt{0.8} - 1 = 11.8\%$. Increasing $C_{L,\max}$ helps reduce stall speed, but a high-lift configuration can also add drag. A lower liftoff speed alone does not prove better climb performance.

What happens when the aircraft must produce more than its weight? Define load factor $n = L/W$. For the same density and configuration,

$$V_{S,n} = \sqrt{\frac{2nW}{\rho S C_{L,\max}}} = V_{S,1}\sqrt{n}.$$

For example, $n = 1.2$ raises the corresponding stall speed by about 9.54%. This is a background relationship, not an additional V-speed rule implemented by V7. It explains why a manoeuvring aircraft can stall at a higher speed.

Why scheduled speeds include margins. VR, V2 and VREF serve different flight phases. Ratios to a reference stall speed provide a simple schedule for a study model. A real schedule must also respect control, handling and other aircraft-specific constraints. V7's ratios are inputs to an approximation; the lift equation alone does not prescribe them.

> **Explain aloud:** stall speed is the speed at which the demanded lift reaches the configuration's available maximum lift coefficient under the stated assumptions.

7 Propeller power, thrust and turbocharging

Theory and further reading. Mair & Birdsall, Ch. 5 [3]; Filippone, Ch. 6 [4]; Gudmundsson [7]. The piecewise thrust cap is the project model.

Power and thrust are different quantities. Useful propulsive power is TV; if propeller efficiency is η_p, then $TV = \eta_p P$ and $T = \eta_p P/V$. V7 limits this relation at low speed:

$$T_{\text{prop}} = \min\left(\sigma_P T_{\text{SLS}}, \frac{\eta_p \sigma_P P_{\max}}{\max(V, 1\,\text{m/s})}\right) f_e.$$

The static-thrust cap prevents the $1/V$ relation predicting infinite thrust at rest. T_{SLS} and $P_{\max}$ are totals for all engines.

$$\sigma_P = \begin{cases} \sigma(h, \Delta\Theta), & \text{normally aspirated,} \\ 1, & \text{turbocharged and } h \le h_{\text{crit}}, \\ \sigma(h, \Delta\Theta)/\sigma(h_{\text{crit}}, \Delta\Theta), & h > h_{\text{crit}}. \end{cases}$$

Below critical altitude, the turbocharged model retains rated power. Above it, power falls with the density ratio relative to the critical-altitude condition. This is a simplified engine model: temperature penalties below critical altitude, mixture control and detailed propeller efficiency maps are not included.

Example: at $V = 40\,\text{m/s}$, $P = 134000\,\text{W}$ and $\eta_p = 0.65$, $\eta_p P/V = 2177.5\,\text{N}$. If static thrust is 2800 N, the power expression limits thrust to 2177.5 N. At 10 m/s it predicts 8710 N, so the static cap limits it to 2800 N. This illustration uses $\sigma_P = 1$ and all engines.

OEI thrust fraction:

$$f_e = \frac{N_{\text{op}}}{N_e}, \qquad N_{\text{op}} = N_e - 1 \quad \text{for OEI.}$$

A twin therefore retains half its modelled total thrust. This assumes equal engines and omits additional failed-engine drag.

Check yourself. Does losing half the thrust halve climb performance? No. Climb depends on excess thrust $T - D$. If drag was already large, the remaining excess can be much less than half or can become negative.

8 Jet thrust, Mach number and flat rating

Theory and further reading. Filippone, Secs. 5.2–5.3 [4]; Mair & Birdsall, Ch. 5 [3]. V7 uses its own simplified thrust fit.

V7 uses this approximate turbofan thrust law:

$$\begin{aligned}\Delta\Theta_{\text{excess}} &= \max(\Delta\Theta_{\text{ISA}} - \Delta\Theta_{\text{flat}}, 0),\\ \sigma_{\text{eff}} &= \sigma(h, \Delta\Theta_{\text{excess}}),\\ M &= V/a(h, \Delta\Theta_{\text{ISA}}),\\ T_{\text{jet}} &= T_{\text{SLS}}\sigma_{\text{eff}}^{0.7}(1 - k_{\text{lapse}}M)f_e.\end{aligned}$$

The Challenger uses $\Delta\Theta_{\text{flat}} = 15°\text{C}$ and $k_{\text{lapse}} = 0.90$. Altitude still affects σ_{eff} below the flat-rating temperature. The actual temperature is used for speed of sound.

Flat rating in this implementation:

- ISA+10 and ISA+15 both give zero excess temperature.
- ISA+25 gives an excess of 10°C, reducing effective density and thrust.
- At zero TAS, $M = 0$: the Mach multiplier is one.
- At nonzero TAS, temperature can still affect thrust through a and M, even below the flat-rating limit. Thus “perfectly constant thrust below ISA+15” is not exact for this model at every speed.

Illustration: take $T_{\text{SLS}} = 65000\,\text{N}$, $\sigma_{\text{eff}} = 1$, $V = 70\,\text{m/s}$ and $a = 340\,\text{m/s}$. Then $M = 0.20588$ and $T = 65000[1 - 0.9(0.20588)] = 52955.9\,\text{N}$ AEO; OEI gives 26477.9 N for a twin.

Reverse thrust during braking:

$$T_{\text{rev}} = \begin{cases} r_{\text{rev}}T_{\text{SLS}}, & \text{enabled and TAS} > 60\,\text{kt},\\ 0, & \text{otherwise.}\end{cases}$$

The code uses TAS for the cutoff and a constant fraction of SLS thrust above it. This is an approximation, not a detailed reverser schedule. The propeller class returns zero reverse thrust.

Check yourself. Why is this not a full engine model? Its density exponent, Mach lapse, flat rating and reverser fraction approximate behaviour without an engine deck.

8 Theory: power and thrust describe different things

Theory and further reading. Mair & Birdsall, Ch. 5 [3]; Filippone, Chs. 5–6 [4]; Gudmundsson [7].

Thrust is a force; power is a rate of doing work. Moving through distance ds with force T aligned with the motion does work $dE = T\,ds$. Dividing by time gives useful propulsive power $P_{\text{useful}} = TV$. For a propeller receiving shaft power P, use efficiency η_p to write

$$TV = \eta_p P, \qquad T = \frac{\eta_p P}{V}.$$

At approximately constant power and efficiency, thrust decreases as speed rises. This helps explain why acceleration can weaken toward the end of a takeoff roll. It does not imply thrust becomes physically infinite at rest.

Why static thrust needs separate treatment. The expression above assumes an efficiency appropriate to the operating point. At $V = 0$, useful aircraft translational power TV is zero even while a propeller produces finite thrust and puts energy into the air. Treating η_p as a nonzero constant down to zero speed breaks the model. V7 avoids the singularity with a lower speed bound and a static-thrust cap. The cap is a practical model choice, not a derivation of propeller aerodynamics.

Why density affects engine output. A normally aspirated piston engine ingests less air mass in thinner air. Less available oxygen constrains the fuel that can be burned effectively, so available power generally falls. V7 approximates the loss with a density ratio. Turbocharging can maintain intake conditions over part of the altitude range; the project's critical altitude marks where its simplified power-holding assumption ends. Real limits depend on the engine and its controls.

A jet needs a different approximation. Jet thrust depends on mass flow, exhaust momentum, pressure forces and operating conditions. A power-divided-by-speed model is not interchangeable with a jet engine model. V7 instead uses a density/Mach fit. Flat rating represents a control or rating limit: cool-day capability can exceed the permitted rating, so a rise in temperature need not immediately reduce rated thrust. It does not remove every altitude or speed effect.

Explain aloud: OEI removes an engine's contribution to total thrust, but climb depends on the thrust remaining *after drag*. Losing half the thrust can remove much more than half the excess thrust.

9 Ground roll: Newton's second law

Theory and further reading. Mair & Birdsall, Ch. 6 [3]; Filippone, Secs. 9.2.1–9.2.2 [4].

Let s be distance along the runway and θ the runway slope:

$$\theta = \arctan(\text{slope}_{\%}/100), \qquad N = \max(W - L, 0).$$

N is the project's approximate wheel load. A more complete inclined-runway balance would use $W\cos\theta - L$; V7 uses $W - L$ for the small-slope approximation.

Acceleration:

$$\boxed{\frac{dV_{\text{GS}}}{dt} = \frac{T - D - \mu_{\text{roll}}N - W\sin\theta}{m}}, \qquad \frac{ds}{dt} = V_{\text{GS}}.$$

Thrust accelerates the aircraft; drag and rolling friction oppose it. An uphill slope subtracts available force. Lift reduces wheel load and therefore rolling friction. The max prevents a negative friction force if computed lift exceeds weight.

Braking:

$$\boxed{\frac{dV_{\text{GS}}}{dt} = -\frac{D + \mu_{\text{brake}}N + T_{\text{rev}} + W\sin\theta}{m}}.$$

An uphill slope assists stopping. A downhill slope has negative $\sin\theta$ and weakens deceleration. Jet ground spoilers set braking $C_L = 0$, which increases wheel load and braking force in this model. Idle thrust is zero.

Wet runway: $\mu_{\text{brake,wet}} = 0.5\mu_{\text{brake,dry}}$. This is a fixed V7 approximation, not a universal tyre–runway law. It affects braking, not the project's rolling coefficient. Therefore wetness alone does not change its AEO takeoff calculation, but does change accelerate-stop.

Illustration: $m = 1200$ kg, $T = 2500$ N, $D = 500$ N, $L = 4000$ N, $\mu_{\text{roll}} = 0.04$, level runway:

$$W = 11767.98\,\text{N}, \quad F_{\text{roll}} = 0.04(11767.98 - 4000) = 310.72\,\text{N},$$

$$a_{\text{GS}} = (2500 - 500 - 310.72)/1200 = 1.4077\,\text{m/s}^2.$$

A 1% uphill slope adds about 117.67 N of resistance, reducing acceleration by about $0.0981\,\text{m/s}^2$.

Check yourself. Why does raising lift help rolling but hurt braking? Both use wheel load: less load reduces unwanted rolling resistance but also reduces useful tyre braking.

9 Theory: wheel load, slope and energy

Theory and further reading. Ground-performance background: Mair & Birdsall, Ch. 6 [3]; Ojha [8].

Resolve forces in the direction of travel. On an uphill runway, gravity has a component $W \sin\theta$ opposing motion. For small slopes, $\theta \approx \tan\theta$ = slope percent/100. Thus a 1% uphill slope adds approximately $0.01W$ of resistance. Dividing by mass gives an acceleration penalty close to $0.01g_0$, independent of mass for that isolated gravity term.

Why wheel load is not always equal to weight. On a level runway, vertical force balance gives $N + L \approx W$ while the gear remains on the ground. This leads to $N \approx W - L$. For an inclined runway, a more detailed normal balance would involve $W\cos\theta$ and the directions of lift and thrust. V7 retains $N = \max(W - L, 0)$ as a small-slope approximation. The maximum prevents unphysical negative tyre load; it does not model liftoff or suspension dynamics by itself.

Rolling resistance and braking both depend on tyre load in this model: $F_{\text{roll}} = \mu_{\text{roll}} N$ and $F_{\text{brake}} = \mu_{\text{brake}} N$. Lift reduces unwanted rolling resistance during acceleration. During braking, the same reduction in tyre load reduces useful braking force. Spoilers help transfer load back to the wheels. Real friction also depends on speed, slip, tyres and runway condition; a single coefficient suppresses these details.

An energy view of the same calculation. For constant mass in the ground frame, Newton's law gives

$$\frac{d}{dt}\left(\tfrac{1}{2}mV_{\text{GS}}^2\right) = F_{\text{net}} V_{\text{GS}}, \qquad \Delta\left(\tfrac{1}{2}mV_{\text{GS}}^2\right) = \int F_{\text{net}}\, ds.$$

With constant braking force F_b, stopping from speed V_0 takes $s = mV_0^2/(2F_b)$. If $F_b = \mu m g_0$ and all other effects are ignored, $s = V_0^2/(2\mu g_0)$. Mass cancels in this special case. That does not mean weight is irrelevant in V7: scheduled speeds, lift, propulsion and force histories change with the operating case.

> **Explain aloud:** twice the stopping speed requires four times the kinetic-energy removal. Runway length can grow rapidly even when speed changes seem modest. The actual brake-energy share also depends on aerodynamic drag, reverse thrust and other forces.

10 Numerical integration: why distance is not one division

Theory and further reading. Filippone, Sec. 9.3 [4], for the performance problem; SciPy documentation [9] for the solver used by V7.

The ground-roll forces vary with TAS. Lift and drag change with V^2, thrust can change with speed, and wheel load changes with lift. Acceleration therefore is not generally constant.

Constant-acceleration check only:

$$s - s_0 = \frac{V_{\mathrm{GS},f}^2 - V_{\mathrm{GS},0}^2}{2a_{\mathrm{GS}}}.$$

This is useful for an order-of-magnitude check, not as a substitute for the project's integration. At constant deceleration magnitude b, stopping distance is $V_{\mathrm{GS},0}^2/(2b)$. A 10% higher initial GS would then require 21% more stopping distance, if b really stayed constant.

A hand-calculation stepping method:

$$V_{\mathrm{GS},i+1} \approx V_{\mathrm{GS},i} + a_i \Delta t,$$
$$s_{i+1} \approx s_i + V_{\mathrm{GS},i} \Delta t.$$

These are forward-Euler updates, included to explain integration. They are not the RK45 algorithm used by V7. For $V_{\mathrm{GS},i} = 20$ m/s, $a_i = 2$ m/s^2 and $\Delta t = 0.5$ s, the new speed is 21 m/s and Euler adds 10 m of distance. Exact constant-acceleration motion would add 10.25 m over that interval.

What V7 actually does: adaptive RK45 advances the state $[s, V_{\mathrm{GS}}]$, using several derivative evaluations to estimate local error and choose a step. The maximum step is 0.5 s, not a fixed step. The maximum interval is 300 s.

$$g_{\mathrm{accel}} = V_{\mathrm{GS}} + V_w - V_{\mathrm{target}},$$
$$g_{\mathrm{stop}} = V_{\mathrm{GS}} - 0.1.$$

Acceleration stops at an upward crossing of the target TAS. Braking stops at a downward crossing of 0.1 m/s GS, the practical stopping threshold.

> Numerical precision is not physical accuracy. Small integration error does not correct an inaccurate drag coefficient, thrust estimate or runway assumption.

Check yourself. What would a step-size study show? Whether the numerical result is stable as the step limit is reduced. It would not validate the aircraft model itself.

10 Theory: integration, events and numerical error

Theory and further reading. Performance equations: Filippone, Sec. 9.3 [4]. V7's solver interface: SciPy [9, 10].

The state contains what we need to advance. For the roll, use $\mathbf{y} = [s, V_{\mathrm{GS}}]^{\mathsf{T}}$. The force calculation returns the derivatives $\mathbf{f} = [V_{\mathrm{GS}}, F_{\mathrm{net}}/m]^{\mathsf{T}}$. Integration repeatedly estimates how these changing derivatives accumulate. The initial conditions are as important as the force equation: accelerating from rest is a different problem from continuing after an engine failure.

Why evaluating the acceleration once is not enough. An Euler step uses the derivative only at the start. As speed rises within the step, aerodynamic forces and sometimes thrust change. Using the old derivative throughout introduces truncation error. Runge–Kutta methods sample intermediate states to better represent that variation.

Connection to the Fortran ground-roll demonstration. For step Δt, classical RK4 uses

$$\begin{aligned}
\mathbf{k}_1 &= \mathbf{f}(\mathbf{y}_i),\\
\mathbf{k}_2 &= \mathbf{f}(\mathbf{y}_i + \tfrac{1}{2}\Delta t\mathbf{k}_1),\\
\mathbf{k}_3 &= \mathbf{f}(\mathbf{y}_i + \tfrac{1}{2}\Delta t\mathbf{k}_2),\\
\mathbf{k}_4 &= \mathbf{f}(\mathbf{y}_i + \Delta t\mathbf{k}_3),\\
\mathbf{y}_{i+1} &= \mathbf{y}_i + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).
\end{aligned}$$

The displayed form is for the time-independent force model. For sufficiently smooth equations, its global error is fourth order in step size. Piecewise thrust caps can reduce the observed convergence order near a switch. V7 uses adaptive RK45 instead, so identical physics need not give bit-for-bit identical numerical outputs.

Why an event is a separate calculation. A step may start below VR and finish above it. Reporting the entire step would overshoot the desired endpoint. Event location finds where the target condition crosses zero. The Fortran example shortens its last RK4 step by bisection; V7 uses the integration library's event handling.

Verification versus validation. Halve the step or tighten tolerances: a stable answer supports numerical convergence. Compare against independent physical data to assess the model. A converged answer can still be wrong if its forces, constants or assumptions are wrong.

11 Rotation, climb and transition to screen height

Theory and further reading. Mair & Birdsall, Chs. 4 and 6 [3]; Raymer [5]; Torenbeek [6]. The chosen arc and constants are V7 approximations.

The AEO takeoff distance is assembled from three pieces:

$$s_{\text{ground}} = s(V_R) + (V_R - V_w)t_{\text{rot}}, \qquad \text{TOD}_{\text{AEO}} = s_{\text{ground}} + s_{\text{air}}.$$

Rotation is modelled at constant V_R: $t_{\text{rot}} = 1$ s for props and 3 s for jets. The takeoff screen height is 50 ft for props and 35 ft for jets.

Steady-climb approximation:

$$q_2 = \tfrac{1}{2}\rho V_2^2, \quad C_{L,\text{climb}} = \frac{W}{q_2 S},$$
$$\sin\gamma = \frac{T(V_2) - D(V_2, C_{L,\text{climb}})}{W},$$
$$\text{gradient}_{\%} = 100\tan\gamma.$$

The along-path balance is $T - D = W\sin\gamma$. The model uses $L \approx W$ to find C_L; an exact steady-climb normal balance would use $L = W\cos\gamma$. A non-positive excess thrust cannot produce the required climb. The code also limits the inverse-sine input to 0.99 as a numerical safeguard.

Angle versus gradient: with $(T - D)/W = 0.10$, $\gamma = \arcsin(0.10) = 5.74°$, while gradient is 10.05%, not 10°. For small angles, $\sin\gamma \approx \tan\gamma \approx \gamma$ in radians. Rate of climb is a vertical speed: ROC $= V\sin\gamma$. Gradient is a distance ratio. V7 reports the air-path gradient; wind changes the gradient measured over the ground, which is not separately reported.

Transition arc:

$$R_{\text{TR}} = \frac{V_{\text{TR}}^2}{g_0(n-1)}, \qquad h_{\text{TR}} = R_{\text{TR}}(1 - \cos\gamma), \quad n = 1.2.$$

Near the bottom of a pull-up, $(n-1)W$ supplies the normal force needed for curvature: $(n-1)mg_0 \approx mV^2/R$. This explains the radius formula.

$$s_{\text{air,still}} = \begin{cases} \sqrt{R_{\text{TR}}^2 - (R_{\text{TR}} - h_{\text{scr}})^2}, & h_{\text{scr}} \le h_{\text{TR}}, \\ R_{\text{TR}}\sin\gamma + \dfrac{h_{\text{scr}} - h_{\text{TR}}}{\tan\gamma}, & h_{\text{scr}} > h_{\text{TR}}. \end{cases}$$

The first case reaches the screen during the arc. The second completes the arc and adds straight climb. V7 then uses $s_{\text{air}} = s_{\text{air,still}}(V_{\text{TR}} - V_w)/V_{\text{TR}}$. This wind correction is an approximation using a representative speed.

Check yourself. Why not use $h_{\text{scr}}/\tan\gamma$ for the entire airborne distance? That assumes the final climb angle is reached immediately at liftoff.

11 Theory: excess thrust, excess power and curved flight

Theory and further reading. Mair & Birdsall, Ch. 4 [3]; Filippone, Ch. 10 [4]. For an accessible derivation see the Embry–Riddle climbing chapter.

A steady climb is a force balance. Assume constant TAS, thrust aligned with the flight path and no wind for the basic geometry. Resolving along and normal to the flight path gives

$$T - D - W\sin\gamma = 0, \qquad L = W\cos\gamma.$$

For small climb angles, $L \approx W$, which is the approximation V7 uses to calculate climb C_L. Rearranging the along-path balance gives $\sin\gamma = (T - D)/W$. The useful quantity is *excess thrust*, $T - D$, not total thrust alone.

Angle and rate answer different questions. Angle describes steepness; rate describes height gained per time. In still air,

$$\text{ROC} = V\sin\gamma = \frac{(T - D)V}{W} = \frac{P_{\text{excess}}}{W}, \qquad \text{gradient} = 100\tan\gamma.$$

At fixed weight, maximum climb angle depends on maximum excess thrust, while maximum climb rate depends on maximum excess power. These need not occur at the same speed. A 5% gradient means 5 m of height per 100 m horizontally; it does not mean a 5-degree climb angle. With wind, ground-referenced gradient also depends on horizontal ground speed; V7's reported angle-based gradient comes from its aerodynamic calculation.

Why engine loss is especially severe. If AEO thrust is 50 kN and drag is 20 kN, excess thrust is 30 kN. Halving thrust leaves 25 kN total but only 5 kN excess: one-sixth of the original. This is why an aircraft can accelerate AEO yet struggle to climb OEI.

Where the transition radius comes from. Changing the flight-path direction requires a normal acceleration V^2/R. Near the start of a pull-up, with thrust-normal effects neglected,

$$L - W \approx mV^2/R, \qquad R \approx \frac{V^2}{g_0(n - 1)}.$$

V7 adopts this near-level relation for a constant-radius transition. Circle geometry then gives height $R(1 - \cos\gamma)$ and horizontal distance $R\sin\gamma$. It is an approximation to the manoeuvre, not a full pitch-response simulation. Raising speed at the same load factor increases the radius as V^2.

> **Explain aloud:** excess thrust sets the possible steady climb angle; excess power sets climb rate; excess normal force curves the flight path.

12 Balanced field length and the AEO limit

Theory and further reading. Mair & Birdsall, Ch. 6 [3]; Filippone, Sec. 9.4 [4]. Root finding follows SciPy [10].

For each trial V_1, assume engine failure occurs exactly at V_1.

$$s_{go}(V_1) = s_{AEO}(0 \to V_1) + s_{OEI}(V_1 \to V_R) + s_{rot} + s_{air,OEI},$$
$$s_{stop}(V_1) = s_{AEO}(0 \to V_1) + (V_1 - V_w)(2\,\text{s}) + s_{brake}.$$

The stop case uses no reverse-thrust credit. The 2 s term is a distance allowance at constant decision-speed GS, not a simulated reaction transient.

An earlier failure generally makes continuing harder, while an earlier rejection leaves less speed and distance to stop from. A later V_1 generally shortens the go case and lengthens the stop case. The balance is the root:

$$f(V_1) = s_{go} - s_{stop} = 0.$$

The search interval is $\max(0.5V_R, V_w + 1\,\text{m/s}) \le V_1 \le V_R$. Brent's method finds a bracketed crossing to a speed tolerance of 0.01 m/s. If the expected crossing is absent, the code selects a bound and labels the case unbalanced. In either case its reported BFL is $\max(s_{go}, s_{stop})$.

The final takeoff requirement can be larger:

$$\boxed{\text{TOFL} = \max(\text{BFL}, 1.15\,\text{TOD}_{AEO})}.$$

> BFL and TOFL are not interchangeable. Balancing go and stop does not remove the separate factored AEO requirement. These are the project's selected distance rules, not a complete certification calculation.

Challenger, sea level, ISA, MTOW, calm, dry:

Quantity	Model result
V_1 / V_R	126.67 / 133.06 KTAS
Accelerate-go / accelerate-stop	4,624.14 / 4,624.14 ft
BFL	4,624.14 ft
AEO TOD / 1.15× AEO TOD	4,199.13 / 4,829.00 ft
TOFL	4,829.00 ft

Check yourself. If BFL is 4500 ft and AEO TOD is 4100 ft, what limits TOFL? $1.15(4100) = 4715$ ft, so the AEO case limits it.

12 Theory: why the balanced-field crossing matters

Theory and further reading. Mair & Birdsall, Ch. 6 [3]; Filippone, Sec. 9.4 [4].

Two possible outcomes share the initial roll. The aircraft first accelerates with all engines. At the selected failure speed, the model branches: continue the takeoff with reduced thrust, or reject it and stop after the prescribed recognition/reaction treatment. Both branches must fit their relevant available runway distances.

Why changing the decision speed moves the curves differently. A higher trial V1 usually helps the continue case because more acceleration occurs AEO before the failure. It usually hurts the reject case because the aircraft reaches a higher speed and travels farther before braking. The go curve therefore often falls while the stop curve rises. These are expected trends for the modeled cases, not a proof of monotonicity for every possible configuration or constraint.

The crossing is a minimax idea. For a trial speed V_1, define the needed field length

$$R(V_1) = \max\{s_{\text{go}}(V_1), s_{\text{stop}}(V_1)\}.$$

If one curve decreases continuously and the other increases continuously, and they cross inside the allowed interval, the maximum of the two is minimized at their crossing. On one side, the go case dominates; on the other, the stop case dominates. This is why balancing can give an efficient decision speed.

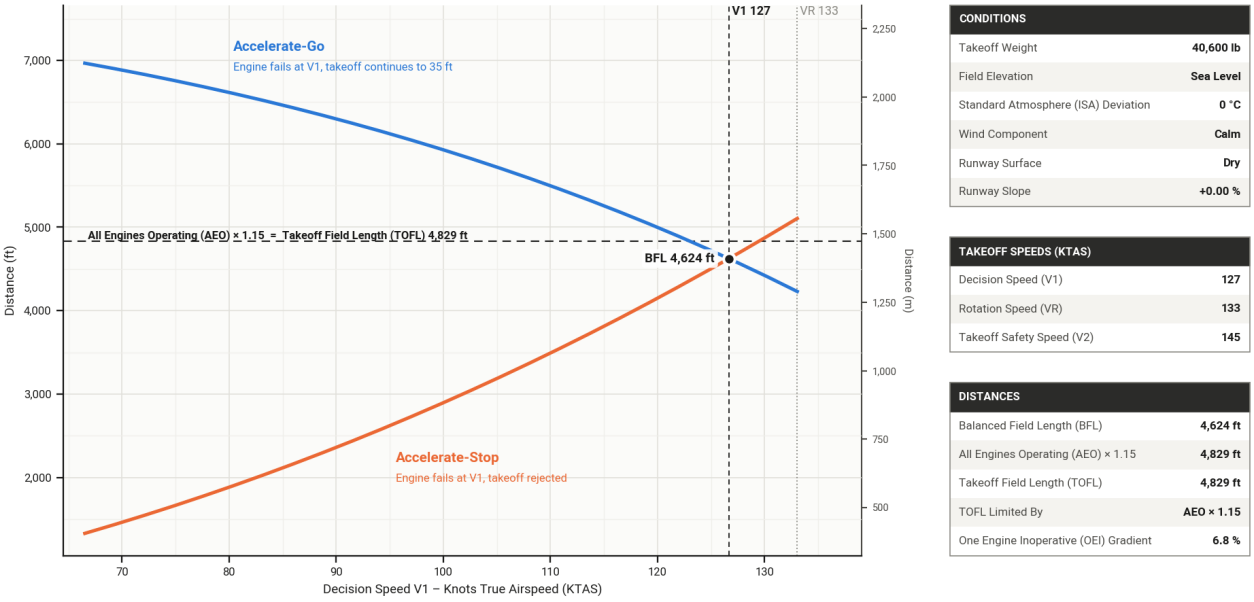
The numerical root problem is $f(V_1) = s_{\text{go}}(V_1) - s_{\text{stop}}(V_1) = 0$. A bracket with opposite signs supplies evidence of a crossing for a continuous function. If there is no valid crossing in the allowed interval, the program cannot simply report a balanced result. A boundary or another constraint may control the problem instead.

Why BFL is not the only required distance. The project's final TOFL also compares the balanced value with its factored AEO case. If the latter is larger, balancing the engine-failure cases does not control the final requirement. Furthermore, a scalar TOFL comparison does not replace the separate ASD versus ASDA check.

> **Explain aloud:** BFL balances two modeled consequences of a failure. It does not prove the aircraft meets every runway, obstacle, brake-energy or control-speed constraint. Each integrated branch and each allowed speed bound must be physically meaningful first.

13 Reading the BFL chart

Challenger 3500 – Balanced Field Length (BFL)



CONDITIONS	
Takeoff Weight	40,600 lb
Field Elevation	Sea Level
Standard Atmosphere (ISA) Deviation	0 °C
Wind Component	Calm
Runway Surface	Dry
Runway Slope	+0.00 %

TAKEOFF SPEEDS (KTAS)	
Decision Speed (V1)	127
Rotation Speed (VR)	133
Takeoff Safety Speed (V2)	145

DISTANCES	
Balanced Field Length (BFL)	4,624 ft
All Engines Operating (AEO) × 1.15	4,829 ft
Takeoff Field Length (TOFL)	4,829 ft
TOFL Limited By	AEO × 1.15
One Engine Inoperative (OEI) Gradient	6.8 %

All speeds are true airspeed. TOFL = max(BFL, 1.15 × AEO takeoff distance), CAR 525 / 14 CFR 25.113.

1. The horizontal axis is trial V_1 in KTAS. The vertical axis is distance. Each point on a curve is a separate continue-or-stop calculation.
2. The descending accelerate-go curve reflects less OEI acceleration when failure occurs later. The rising accelerate-stop curve reflects a later rejection.
3. Their crossing gives the balanced decision speed and BFL. Do not choose V_1 beyond V_R, even if a plotted extrapolation would appear favourable.
4. Compare the crossing height with the horizontal factored AEO distance. The larger distance sets TOFL. For this baseline Challenger case, the AEO line sits above the BFL crossing.
5. A small difference between the two limiting distances means a change in conditions may change which case governs. Always calculate both.

Check yourself. If the stop curve moves upward because braking friction decreases, which way would the crossing normally move? Toward a lower V_1, provided the go curve is unchanged and an intersection still exists within the bounds.

14 Landing distance and landing-distance factors

Theory and further reading. Mair & Birdsall, Ch. 6 [3]; Filippone, Sec. 11.7 [4]. Operational factors are separate from textbook force balances.

Landing starts at a 50 ft screen height for both aircraft types:

$$V_{REF} = 1.3V_{S0}, \qquad V_{TD} = 1.15V_{S0},$$
$$s_{air} = \frac{h_{scr}}{\tan 3^\circ}\frac{V_{REF} - V_w}{V_{REF}},$$
$$s_{free} = (V_{TD} - V_w)t_{free},$$
$$LD = s_{air} + s_{free} + s_{brake}.$$

t_{free} is 2 s for props and 3 s for jets. It represents travel after touchdown before full braking; the speed is held constant during this segment. Braking starts at $V_{TD} - V_w$ GS and uses the braking force balance. The lower touchdown speed approximates speed lost during the flare; the flare itself is not integrated as a separate flight path.

Geometry check: 50 ft = 15.24 m, so the still-air distance on a 3° approach is $15.24/\tan 3^\circ = 290.80$ m. With a 60 m/s approach TAS and 5 m/s headwind, the model reduces this to $290.80(55/60) = 266.57$ m.

LD versus LDR. LD is the calculated physical landing distance. For jets, runway analysis uses:

$$LDR_{dry} = \frac{LD_{dry}}{0.6}, \qquad LDR_{wet} = 1.15\,LDR_{dry}.$$

For a dry LD of 800 m, LDR is 1333.33 m dry and 1533.33 m wet. Dividing by 0.6 increases the requirement; it does not reduce the distance.

> Two different wet effects exist in V7. A wet physical landing uses half the braking coefficient. The wet jet runway requirement is separately calculated from the *dry* landing distance and multiplied by 1.15. Do not apply 1.15 to the wet physical LD and call it the same calculation.

The standalone landing result's LDR field is LD times the aircraft factor for the supplied conditions. The runway-analysis function performs the separate dry-baseline recalculation needed for its wet jet requirement. Props have no LDR factor in this model; their runway check uses unfactored LD.

Check yourself. If reverse thrust reduces LD, will the default runway report use that benefit? No. Runway analysis defaults to reverse thrust disabled.

14 Theory: landing geometry and stopping energy

Theory and further reading. Mair & Birdsall, Ch. 6 [3]; Filippone, Sec. 11.7 [4].

Landing distance contains different physical processes. The airborne segment loses height, the free-roll segment travels before full braking, and the braking segment removes ground-frame kinetic energy. Adding the three distances is reasonable only if the end condition of each segment matches the start condition of the next.

Approach geometry is a triangle approximation. At constant descent angle γ in still air, the horizontal distance to lose height h is $s = h/\tan|\gamma|$. For $h = 15.24$ m and $|\gamma| = 3°$, $s \approx 290.80$ m. A smaller angle requires more horizontal distance. This simple construction does not independently resolve a flare, touchdown sink rate or a changing flight path. V7 uses a fixed geometry with a wind correction; learn that approximation rather than interpreting it as a trajectory simulation.

Delay costs distance even before braking starts. At approximately constant touchdown GS, $s_{\text{free}} = V_{\text{GS}} t_{\text{free}}$. At 30 m/s, each additional second adds about 30 m before the braking calculation. Then a simplified constant-deceleration estimate gives $s_{\text{brake}} = V_0^2/(2b)$. This explains why touchdown speed and delay can matter as much as braking strength. In V7, forces change during braking, so integration replaces that simple estimate.

Separate a physical prediction from an operating allowance. Physical LD estimates the distance taken by the modeled landing. Factored LDR is a requirement used for a runway comparison. A multiplier changes the required distance; it does not change the simulated force balance or add real brake force. The project's dry and wet factors belong to this second category.

Wetness appears in two distinct ways: V7 can reduce the braking coefficient in a physical wet landing, and its jet runway analysis can apply a wet multiplier to the dry-based factored requirement. Combining these without following the specified calculation would answer a different question.

> **Explain aloud:** the landing model joins a geometry estimate, a delay estimate and a braking trajectory. Operational factors are applied to a defined baseline after the physical calculation; they are not extra forces.

15 Runway checks, margins and sensitivity

Theory and further reading. Ojha [8] and Filippone, Ch. 9 [4], for performance context. The margin definitions and scenario choices below follow V7.

Abbreviation	Available distance
TORA	Takeoff Run Available: ground run.
TODA	Takeoff Distance Available: takeoff distance including permitted clearway.
ASDA	Accelerate-Stop Distance Available: stopping distance including permitted stopway.
LDA	Landing Distance Available.

For built-in runways, V7 simplifies TORA = TODA = ASDA = runway length and LDA = runway length minus the displaced threshold. It does not model actual clearway/stopway declarations independently for those defaults.

$$\begin{aligned} \text{Twin takeoff:} &\quad \text{TOFL} \le \text{TODA}, \quad \text{ASD} \le \text{ASDA}, \\ \text{Single takeoff:} &\quad \text{TOD} \le \text{TODA}, \quad s_{\text{ground}} \le \text{TORA}, \\ \text{Jet landing:} &\quad \text{LDR} \le \text{LDA}, \\ \text{Prop landing:} &\quad \text{LD} \le \text{LDA}. \end{aligned}$$

$$\text{margin} = \text{available} - \text{required}.$$

A zero margin passes the implemented comparison. A negative margin fails it. Passing these checks does not prove obstacle clearance, brake-energy compliance, OEI second-segment climb compliance or an acceptable crosswind.

KICT 01L example. Challenger at MTOW/MLW, OAT 30°C, reported wind 020°/10 kt, dry runway: ISA deviation is +17.64°C and performance headwind is 5.00 kt. Available distances are 10302 ft. TOFL is 5,107.98 ft; ASD is 4,896.48 ft; LDR is 4,429.66 ft. The landing margin is therefore 5,872.34 ft.

Sensitivity calculation:

$$\Delta_{\%} = 100 \left(\frac{s}{s_{\text{baseline}}} - 1 \right).$$

Changing distance from 1500 to 1800 m gives +20%; changing it to 1200 m gives −20%. The sweeps change one input at a time; takeoff weight is a percentage of MTOW, landing weight a percentage of MLW. A sweep is not a combined hot/high, heavy, tailwind worst-case assessment.

Check yourself. Why can two aircraft have the same percentage weight setting but different masses? Their MTOW and MLW are different reference values.

16 A complete C172 calculation to explain aloud

Inputs: sea level, ISA, calm wind, dry level runway, MTOW = MLW = 2550 lb; $S = 16.2\ \text{m}^2$, $C_{L,\max,TO} = 1.5$, $C_{L,\max,LDG} = 2.0$, $C_{L,\text{gnd}} = 0.4$, $C_{D_0} = 0.032$, $k = 0.045$, $\mu_{\text{roll}} = 0.04$, $\mu_{\text{brake}} = 0.40$.

1. Convert mass: $m = 1,156.66$ kg and $W = 11,342.97$ N.
2. Calculate the atmosphere: $\rho = 1.225012\ \text{kg/m}^3$ and $a = 340.29$ m/s. The density is slightly different from exactly 1.225 because the stored gas constant and sea-level density reference are rounded independently.
3. Derive speeds: $V_{S1} = 27.61$ m/s, $V_R = 30.37$ m/s, $V_2 = 33.13$ m/s.
4. At V_R, calculate $q = 564.81$ Pa, $L = 3,660.00$ N, $D = 358.68$ N and $T = 2,800.03$ N.
5. Rolling friction is 307.32 N, giving instantaneous acceleration $1.84\ \text{m/s}^2$. This is one point on the trajectory, not the average.
6. Integrate acceleration to V_R, then add 30.37 m for rotation. Total ground distance is 268.61 m, including rotation; total ground time is 16.45 s.
7. Use excess thrust at V_2 to obtain a climb gradient of 15.61%. The transition/climb calculation adds 137.51 m to the 50 ft screen.
8. TOD is 268.61 + 137.51 = 406.11 m, or 1,332.40 ft (the displayed rounded components may differ slightly in their last digit).
9. For landing, use $V_{\text{REF}} = 31.08$ m/s and $V_{\text{TD}} = 27.49$ m/s. The air, free-roll and braking distances are 290.80, 54.99 and 107.48 m.
10. LD is 453.26 m, or 1,487.09 ft. No jet LDR factor is applied.

> You should be able to explain where every number in this chain comes from. The ground-roll and braking totals require integration; they cannot be reproduced exactly by substituting one average acceleration into a closed-form equation.

17 What changes what, and why

Change	Main reasoning to explain
Higher pressure altitude	Density usually falls; stall and scheduled TAS rise. Engine thrust/power may fall. More GS and less excess force tend to increase distance.
Higher OAT	Density falls at fixed pressure altitude. Engine response depends on the turbocharging or flat-rating approximation.
Higher mass	Weight and required lift rise; stall speed rises as $\sqrt{W}$. There is more kinetic energy at a given GS and less acceleration for a given net force.
Headwind	Target TAS is reached at lower GS. The ground-distance and airborne wind corrections generally reduce the required distance.
Tailwind	Higher GS is needed for the same TAS; distance generally increases. The runway-analysis factor magnifies the input penalty.
Uphill slope	Reduces acceleration, assists braking. Downhill reverses those effects.
Wet runway	Weaker tyre braking increases stopping distance. V7 does not implement the full wet takeoff treatment.
Ground spoilers	Lower lift puts more weight on the wheels and increases braking force.
Higher $C_{L,\max}$	Lowers stall speed and the scheduled speeds. It does not by itself specify the real drag cost of a flap change.

Calibration versus validation. Calibration adjusts an input to reproduce a chosen target. V7 calibrates Challenger takeoff $C_{L,\max}$ to TOFL and SR22T static thrust to takeoff ground roll. Matching those targets is not independent validation. Landing distances or other conditions not used for tuning provide separate comparisons, with their own data uncertainty.

Know the main omissions: QNH correction; IAS/CAS conversion; a full engine deck; failed-engine drag; VMCG/VMU schedules; second-segment OEI/WAT limits; brake-energy limits; complete wet-runway certification; verified declared runway distances. These explain why a plausible numerical result can still be incomplete.

Check yourself. What would you improve first? State a specific missing effect, the data needed to model it, and an independent comparison that would test the change.

17 Theory: what a sensitivity result can and cannot prove

Theory and further reading. Performance context: Ojha [8]; the distinctions below apply to the V7 scenario calculations.

A calculation answers a conditional question. A model output means: with these inputs, coefficients and assumptions, the equations predict this result. Changing one input while holding others fixed shows a conditional sensitivity. It does not isolate every effect that would change together in a real aircraft operation.

Separate absolute change from percentage change. For a baseline s_0 and changed case s_1,

$$\Delta s = s_1 - s_0, \qquad \Delta s_{\%} = 100\frac{s_1 - s_0}{s_0}.$$

An additional 100 m is a 20% increase from 500 m but only 5% from 2000 m. Report the reference condition with every percentage. A sweep over many inputs is not equivalent to a local derivative at one condition.

Use a small perturbation to estimate local sensitivity. For an input x and a small increment δx, a central difference is

$$\frac{\partial s}{\partial x} \approx \frac{s(x + \delta x) - s(x - \delta x)}{2\delta x}.$$

The units matter: metres per kilogram differs from metres per degree Celsius. Very small perturbations can expose numerical noise; large ones average over nonlinear changes. A switch between BFL-controlled and AEO-controlled TOFL can also create a corner where a single smooth derivative is misleading. This derivative is an optional analysis technique, not a claim that V7's existing percentage sweep uses it.

Calibration does not supply independent evidence. Adjusting static thrust to match one measured roll can make that example agree by construction. Other parameter combinations might match it too. Assess the model against conditions not used for tuning: different weights, temperatures, altitudes or speeds. Otherwise, you cannot tell whether the fitted parameter captures the intended physics or compensates for another missing effect.

A runway margin is only as strong as the requirement behind it. Margin is available distance minus required distance. Positive margin means the modeled comparison passes. It does not account for omitted obstacles, incorrect inputs or unmodeled operational constraints. Greater numerical precision cannot remove uncertainty in friction, thrust or maximum lift coefficient.

> **Explain aloud:** verification asks whether the equations were solved correctly; validation asks whether those equations describe the intended physical case well enough. Calibration fits parameters; sensitivity measures response. These are related tasks, but none substitutes for the others.

18 Practice questions: try these without the answers

Use $g_0 = 9.80665$ m/s^2; round only at the end.

1. Convert 100 kt to m/s and 5000 ft to m. Convert 3000 lb to mass and weight in SI.
2. At 1500 m pressure altitude and OAT 25°C, find standard temperature in K and °C, ISA deviation and actual temperature in K.
3. For $p = 90000$ Pa and $\Theta = 300$ K, calculate density and σ.
4. Wind is from 060° at 20 kt on a runway heading 030°. Find headwind, crosswind and performance headwind.
5. With $\rho = 1.1$ kg/m^3, $V = 50$ m/s, $S = 20$ m^2 and $C_L = 0.6$, find q and lift. With $C_{D_0} = 0.03$ and $k = 0.05$, find drag.
6. For $W = 20000$ N, $\rho = 1.1$, $S = 20$ and $C_{L,\max} = 1.6$, calculate V_{S1}, V_R and V_2 in m/s.
7. If weight rises by 21% with density and $C_{L,\max}$ unchanged, what is the percentage increase in stall speed?
8. For $P = 200000$ W, $\eta_p = 0.75$, $V = 30$ m/s, $T_{\mathrm{SLS}} = 4000$ N and $\sigma_P = 1$, find propeller thrust with all engines.
9. A twin has AEO thrust 50000 N, drag 20000 N and weight 200000 N. Compare AEO and OEI climb gradients using the project's steady-climb relations.
10. An aircraft reaches 40 m/s GS from rest at a constant 2 m/s^2. Find time and distance. Explain why this is only a check for the real model.
11. At $V_R = 60$ m/s TAS with 5 m/s headwind and a 3 s rotation, what rotation distance is added?
12. Let $V_{\mathrm{TR}} = 60$ m/s, $n = 1.2$ and $\gamma = 0.10$ rad. Find R_{TR} and h_{TR}. Does a 10 m screen lie within the arc?
13. BFL is 1400 m and AEO TOD is 1300 m. Find TOFL and the limiting case.
14. Dry jet LD is 900 m and LDA is 1700 m. Find dry and wet LDR and margins.
15. Distance rises from 1600 to 1840 m. Find the percentage change. If TOFL passes TODA but ASD exceeds ASDA, does takeoff pass?

19 Worked answers

1. 100(1852/3600) = 51.4444 m/s; 5000(0.3048) = 1524 m. Mass = 3000(0.45359237) = 1360.7771 kg; weight = 13344.6648 N.
2. Θ_{std} = 288.15 − 0.0065(1500) = 278.40 K, or 5.25°C. ISA deviation = 25 − 5.25 = 19.75°C. Actual temperature = 298.15 K.
3. $\rho = 90000/(287.05 \times 300) = 1.0451$ kg/m^3; $\sigma = \rho/1.225 = 0.8532$.
4. Heading difference = 30°. Headwind = 17.3205 kt, crosswind = 10 kt from the right; performance headwind = 8.6603 kt.
5. $q = 0.5(1.1)(50)^2 = 1375$ Pa; $L = 1375(20)(0.6) = 16500$ N. $C_D = 0.03 + 0.05(0.6)^2 = 0.048$; $D = 1320$ N.
6. $V_{S1} = \sqrt{40000/(1.1 \times 20 \times 1.6)} = 33.7100$ m/s; $V_R = 37.0810$ m/s; $V_2 = 40.4520$ m/s.
7. Speed factor = $\sqrt{1.21} = 1.10$: a 10% increase.
8. Power-based thrust = 0.75(200000)/30 = 5000 N. Static cap = 4000 N, so thrust is 4000 N.
9. AEO: $\sin\gamma = (50000 - 20000)/200000 = 0.15$; gradient = $100\tan(\arcsin 0.15) = 15.17\%$. OEI thrust = 25000 N, giving $\sin\gamma = 0.025$ and gradient = 2.501%. Half the thrust leaves only one-sixth of the original excess thrust.
10. $t = 40/2 = 20$ s; $s = 40^2/(2 \times 2) = 400$ m. Actual thrust, drag and friction vary during the ground roll.
11. $s_{\text{rot}} = (60 - 5)(3) = 165$ m.
12. $R_{\text{TR}} = 60^2/[9.80665(0.2)] = 1835.49$ m; $h_{\text{TR}} = 1835.49(1 - \cos 0.10) = 9.17$ m. The 10 m screen is above the completed arc, so add a short straight-climb segment.
13. 1.15(1300) = 1495 m, larger than BFL. TOFL = 1495 m, AEO-limited.
14. Dry LDR = 900/0.6 = 1500 m, margin = +200 m. Wet LDR = 1.15(1500) = 1725 m, margin = −25 m: wet requirement fails.
15. 100(1840/1600 − 1) = 15%. The takeoff check fails if ASD exceeds ASDA, even if TOFL fits within TODA.

20 Questions you should be able to answer aloud

1. **Why use TAS for forces and GS for distance?** Air loads depend on relative airflow; runway travel depends on motion relative to the ground.
2. **Why does hot-and-high operation need more runway?** Lower density raises the scheduled TAS and can reduce propulsion. The size of each effect depends on the engine model and the particular case.
3. **Why not use a single acceleration value?** The force balance changes with speed, especially thrust, aerodynamic loads and wheel friction.
4. **Why can spoilers improve stopping?** They reduce lift, increase wheel load and increase the available tyre braking force in the model.
5. **What is BFL?** The go/stop distance at their crossing, when a valid crossing exists within the selected decision-speed bounds.
6. **Why might TOFL exceed BFL?** The factored AEO distance may be larger.
7. **What makes a root-finding method different from an integrator?** Integration predicts a trajectory for given conditions. Root finding changes V_1 until two integrated distance outcomes agree.
8. **What does a calibration match prove?** That a selected parameter can reproduce that target, not that the complete model is independently validated.
9. **Why are the fixed speed ratios limited?** Real schedules also depend on configuration, control speeds, certification constraints and manufacturer data.
10. **Which wet landing distance is factored?** Runway analysis uses the dry jet LD for its LDR factor and wet multiplier, separately from wet physical braking.
11. **What does a positive runway margin prove?** Only that the specified model requirement fits the supplied available distance under the selected assumptions.
12. **How would you test an improvement?** Use data not used to tune it, compare several operating conditions, and separate numerical error from model error.

You are ready to explain the project when you can reconstruct the C172 example, explain the BFL chart, solve the practice set and identify the main limitations without reading a prepared script.

21 Compact equation checklist

$$W = mg_0, \qquad q = \tfrac{1}{2}\rho V^2,$$
$$\rho = p/(R\Theta), \qquad a = \sqrt{1.4R\Theta},$$
$$L = qSC_L, \qquad D = qS(C_{D_0} + kC_L^2),$$
$$V_S = \sqrt{2W/(\rho S C_{L,\max})}, \qquad V_{\mathrm{GS}} = V_{\mathrm{TAS}} - V_w,$$
$$\dot{V}_{\mathrm{GS,accel}} = (T - D - \mu_{\mathrm{roll}}N - W\sin\theta)/m,$$
$$\dot{V}_{\mathrm{GS,brake}} = -(D + \mu_{\mathrm{brake}}N + T_{\mathrm{rev}} + W\sin\theta)/m,$$
$$\sin\gamma = (T - D)/W, \qquad G_{\%} = 100\tan\gamma,$$
$$R_{\mathrm{TR}} = V_{\mathrm{TR}}^2/[g_0(n-1)], \qquad h_{\mathrm{TR}} = R_{\mathrm{TR}}(1 - \cos\gamma),$$
$$\mathrm{TOD} = s_{\mathrm{roll}} + s_{\mathrm{rot}} + s_{\mathrm{air}},$$
$$\mathrm{TOFL} = \max(\mathrm{BFL}, 1.15\mathrm{TOD}_{\mathrm{AEO}}),$$
$$\mathrm{LD} = s_{\mathrm{air}} + s_{\mathrm{free}} + s_{\mathrm{brake}},$$
$$\mathrm{LDR}_{\mathrm{dry}} = \mathrm{LD}_{\mathrm{dry}}/0.6, \qquad \mathrm{LDR}_{\mathrm{wet}} = 1.15\mathrm{LDR}_{\mathrm{dry}}.$$

Here $N = \max(W - L, 0)$ is the project's wheel-load approximation. Keep the detailed piecewise propulsion, wind and transition equations from the earlier sections.

Source of the project-specific equations and values. The current V7 files `atmosphere.py`, `aircraft.py`, `segments.py`, `takeoff.py`, `bfl.py`, `landing.py`, `airports.py`, `runway_analysis.py`, `sensitivity.py` and `units.py`. The methodology report and `REFERENCES.md` give the project's textbook and regulatory references. The factors here describe the implementation; this guide is not a current regulatory compliance review.

Additional explanations checked for this guide:

- NASA Glenn, Lift Equation and Drag Equation: definitions of aerodynamic force and coefficients.
- NASA Glenn, Induced Drag Coefficient: finite-wing interpretation of k.
- NASA Glenn, Earth Atmosphere Equation: atmosphere-model background. Its curve-fit constants differ from V7; use V7's constants for matching calculations.

22 Textbook reading map

Use the topic citations to connect a calculation to its physical background. For a first pass, read Mair & Birdsall alongside the force balances. Use Filippone for a more detailed treatment of field performance and engine models.

Topic	Mair & Birdsall [3]	Filippone [4]
Aerodynamics	Chs. 1–3	Ch. 4
Atmosphere and speeds	Ch. 2 (flight theory)	Secs. 8.1–8.2
Propulsion	Ch. 5	Chs. 5–6
Ground run	Ch. 6	Secs. 9.2–9.3
OEI and stopping	Ch. 6	Sec. 9.4
Climb	Ch. 4	Ch. 10
Landing	Ch. 6	Sec. 11.7

Other useful textbooks in the project bibliography. Anderson's two books [1, 2] support the fundamentals. Raymer [5] and Torenbeek [6] put performance estimates in the wider aircraft-design process. Gudmundsson [7] is especially relevant to general aviation; Ojha [8] provides another flight-performance treatment. These are book-level recommendations, without claims about specific pages or identical implementations.

Keep the source categories separate. Newton's law and aerodynamic definitions supply the physical framework. The code supplies the numerical constants, piecewise engine fits, fixed speed ratios, wet-braking reduction and integration settings used in these examples. Regulations supply requirements in their stated scope; a simplified implementation of a factor does not reproduce an entire certification or operating rule.

Checking the reading pointers. The publisher's contents listing for Mair & Birdsall and front matter for Filippone identify the chapters and sections above. These checks establish where to read, not a page-by-page verification of the textbooks' equations.

Regulatory reading recorded by V7. The project cites 14 CFR 25.105, 25.109 and 25.113 for takeoff context and 14 CFR 121.195 for landing limitations. Consult the applicable rule and aircraft data before interpreting these factors outside the educational model.

References

[1] Anderson, John D., Jr. *Aircraft Performance and Design*. McGraw-Hill, 1999.

[2] Anderson, John D., Jr. *Introduction to Flight*. 8th ed., McGraw-Hill Education, 2016.

[3] Mair, W. Austyn, and David L. Birdsall. *Aircraft Performance*. Cambridge University Press, 1992. doi:10.1017/CBO9780511607134.

[4] Filippone, Antonio. *Advanced Aircraft Flight Performance*. Cambridge University Press, 2012 (print edition). doi:10.1017/CBO9781139161893.

[5] Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 6th ed., American Institute of Aeronautics and Astronautics, 2018.

[6] Torenbeek, Egbert. *Synthesis of Subsonic Airplane Design*. Delft University Press / Kluwer Academic Publishers, 1982. doi:10.1007/978-94-017-3202-4.

[7] Gudmundsson, Snorri. *General Aviation Aircraft Design: Applied Methods and Procedures*. 1st ed., Butterworth-Heinemann, 2014.

[8] Ojha, S. K. *Flight Performance of Aircraft*. American Institute of Aeronautics and Astronautics, 1995.

[9] SciPy developers. `scipy.integrate.solve_ivp`, API documentation. Solver methods, tolerances and events. Accessed 4 November 2026.

[10] SciPy developers. `scipy.optimize.brentq`, API documentation. Bracketed scalar root finding. Accessed 4 November 2026.